# Switching Loss

## Introduction

In the rapidly evolving landscape of charging infrastructure and power electronics, the demand for smaller, lighter, and more efficient power converters is paramount. Whether it's an onboard charger for an electric vehicle or a compact power supply, reducing the physical footprint and weight of these devices is a key design objective. This drive towards miniaturization often leads engineers to increase the operating frequency of power converters, specifically the switching frequency. While higher switching frequencies offer significant advantages in reducing the size of passive components like inductors, capacitors, and transformers, they introduce a critical challenge: increased switching losses within the semiconductor switches. Understanding these losses is fundamental, as they directly impact the overall efficiency, thermal management, and ultimately, the reliability of the power converter. This lecture delves into the mechanisms behind switching losses, the factors that influence them, and why they are a crucial consideration in the design of modern DC-DC converters, particularly isolated topologies like the full-bridge converter.

## Core Concepts

| Term | Definition | Significance |
|---|---|---|
| `Switching Frequency (fsw)` | The rate at which a power semiconductor switch turns on and off in a converter circuit. | Higher frequencies enable smaller passive components, reducing converter size and weight, but increase switching losses. |
| `Switching Loss` | Energy dissipated in a semiconductor switch during its transition between ON and OFF states. | A major contributor to overall power converter inefficiency, especially at high switching frequencies. |
| `Passive Components` | Non-active electrical components like inductors, capacitors, and transformers, | Their size and weight are inversely proportional to switching frequency, making |

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| | crucial for energy storage and filtering. | them targets for miniaturization. |
| `Turn-on Time (T_switch_on)` | The total time duration during which a switch transitions from its OFF state to its ON state, encompassing current rise and voltage fall. | A shorter turn-on time generally leads to lower turn-on switching losses. |
| `Turn-off Time (T_switch_off)` | The total time duration during which a switch transitions from its ON state to its OFF state, encompassing voltage rise and current fall. | A shorter turn-off time generally leads to lower turn-off switching losses. |
| `Blocking Voltage` | The voltage across a switch when it is in its OFF state, preventing current flow. | Higher blocking voltages during switching transitions contribute to increased switching losses. |
| `Switching Current` | The current flowing through a switch during its transition between ON and OFF states. | Higher currents during switching transitions contribute to increased switching losses. |

## Detailed Analysis

### 1. The Rationale for High Switching Frequency and its Consequence

The primary motivation for operating DC-DC converters at high switching frequencies (fsw) stems from the desire to reduce the size and weight of the overall power converter. This is particularly critical for applications like charging infrastructure, where space and weight are often at a premium, such as in onboard vehicle chargers.

- **Reduction in Passive Component Size**:
- **Transformer**: The cross-sectional area of a transformer's core is inversely proportional to the switching frequency. By increasing fsw, a smaller core

can be used for the same power transfer, leading to a more compact and lighter transformer.

- **Inductors and Capacitors**: Filter inductors and capacitors, essential for smoothing output voltage and current, also exhibit an inverse relationship between their required values and the switching frequency. Higher fsw allows for smaller inductance and capacitance values, further reducing the size and weight of these passive components.
- **Overall Converter Size and Weight**: Since the size of power converters is largely dominated by these passive components (inductors, capacitors, and high-frequency transformers), increasing fsw directly leads to a reduction in the converter's volume and mass.
- **The Trade-off**: While beneficial for miniaturization, increasing the switching frequency comes with a significant drawback: an increase in switching losses within the semiconductor switches. Each time a switch turns on or off, energy is dissipated, and at higher frequencies, these loss events occur more frequently, leading to higher total power loss.

### 2. Analyzing Switching Losses in a Buck Converter

To understand the mechanics of switching losses, a simplified buck converter topology is often used as a representative model. This model typically consists of a main switch (S1) and a freewheeling diode (D), or a synchronous rectifier (S2) in place of the diode. For analysis, the output inductor current (IL) is often approximated as a constant current (I0), assuming a very small ripple. This simplification allows for a clear focus on the voltage and current transitions within the switch.

- **Assumptions for Simplification**:
- Ideal switch characteristics (zero voltage drop when ON, zero current when OFF).
- Negligible forward voltage drop of the diode.
- No consideration of non-linear input/output capacitances of the devices.
- Output inductor current (IL) is approximated as a constant I0, neglecting ripple for initial analysis.

### 3. Energy Loss During Switch Turn-on

When a switch transitions from its OFF state to its ON state, both voltage across it and current through it undergo significant changes, leading to energy dissipation.

- **Current Rise Period (TRI)**:

- Initially, the switch is OFF, blocking the input voltage (Vin), and carrying no current.
- Upon receiving a turn-on signal, the current through the switch (IS1) begins to rise from zero towards the load current (I0).
- During this period, the voltage across the switch (VS1) remains high, approximately equal to Vin, because the freewheeling diode is still conducting the load current, keeping the switch's output node at a low potential.
- Energy loss during TRI is calculated by integrating the product of instantaneous voltage and current over this time. It is found to be proportional to Vin, I0, and TRI.
- **Voltage Fall Period (TFV)**:
- Once the switch current (IS1) reaches the load current (I0), the freewheeling diode becomes reverse-biased and stops conducting.
- At this point, the voltage across the switch (VS1) begins to fall from Vin towards zero.
- During this period, the current through the switch is approximately I0.
- Energy loss during TFV is calculated by integrating the product of instantaneous voltage and current over this time. It is found to be proportional to Vin, I0, and TFV.
- **Total Turn-on Energy and Power Loss**:
- The total energy loss during one turn-on event (E_on) is the sum of energy losses during TRI and TFV. This total energy is proportional to Vin, I0, and the total turn-on time (T_switch_on = TRI + TFV).
- The average power loss during turn-on (P_switch_ON) is obtained by multiplying E_on by the switching frequency (fsw), as this event occurs once per switching cycle. Thus, P_switch_ON is proportional to Vin, I0, T_switch_on, and fsw.

### 4. Energy Loss During Switch Turn-off

Similarly, when a switch transitions from its ON state to its OFF state, energy is dissipated due to overlapping voltage and current.

- **Voltage Rise Period (TRV)**:
- Initially, the switch is ON, carrying the load current (I0), and having near-zero voltage across it.
- Upon receiving a turn-off signal, the voltage across the switch (VS1) begins to rise from zero towards Vin.

- During this period, the current through the switch (IS1) remains approximately I0, as the freewheeling diode has not yet started conducting.
- Energy loss during TRV is calculated by integrating the product of instantaneous voltage and current over this time. It is found to be proportional to Vin, I0, and TRV.
- **Current Fall Period (TFI)**:
- Once the voltage across the switch (VS1) reaches Vin, the freewheeling diode becomes forward-biased and starts conducting the load current.
- At this point, the current through the switch (IS1) begins to fall from I0 towards zero.
- During this period, the voltage across the switch is approximately Vin.
- Energy loss during TFI is calculated by integrating the product of instantaneous voltage and current over this time. It is found to be proportional to Vin, I0, and TFI.
- **Total Turn-off Energy and Power Loss**:
- The total energy loss during one turn-off event (E_off) is the sum of energy losses during TRV and TFI. This total energy is proportional to Vin, I0, and the total turn-off time (T_switch_off = TRV + TFI).
- The average power loss during turn-off (P_switch_OFF) is obtained by multiplying E_off by the switching frequency (fsw), as this event also occurs once per switching cycle. Thus, P_switch_OFF is proportional to Vin, I0, T_switch_off, and fsw.

**5. Total Switching Power Loss and Influencing Factors**

The total switching power loss (P_switching) in a converter is the sum of the turn-on and turn-off power losses.

- **P_switching = P_switch_ON + P_switch_OFF**.
- This total loss is directly proportional to the switching frequency (fsw), the input voltage (Vin), the load current (I0), and the sum of the turn-on and turn-off times (T_switch_on + T_switch_off).
- **Factors Affecting Switching Losses**:
- **Switching Frequency (fsw)**: Directly proportional; higher fsw means more switching events and thus higher losses.
- **Blocking Voltage (Vin)**: The voltage the switch must block when OFF. Higher blocking voltage during transitions increases losses.
- **Switching Current (I0)**: The current the switch must carry when ON. Higher current during transitions increases losses. (Note: For precise

calculations, the ripple in the inductor current, delta IL, should be considered, meaning the current at turn-on might be I0 - delta IL/2 and at turn-off I0 + delta IL/2).

- **Turn-on and Turn-off Times (T_switch_on, T_switch_off)**: These represent the speed of the switch transitions. Shorter transition times lead to less overlap between high voltage and high current, thus reducing losses. These times are device-dependent and influenced by gate drive characteristics.
- **Strategies for Loss Reduction (Future Concepts)**:
- The lecture hints at advanced techniques to reduce switching losses. If the voltage across the device can be brought to zero before it turns on (Zero Voltage Switching - ZVS), or if the current through the device can be brought to zero before it turns off (Zero Current Switching - ZCS), the switching losses can be significantly reduced, ideally to zero. These "soft switching" techniques are crucial for high-frequency, high-efficiency converters.

## Key Takeaways

- [ ] Higher switching frequency reduces the size and weight of passive components (inductors, capacitors, transformers).
- [ ] Increased switching frequency directly leads to higher switching losses in semiconductor switches.
- [ ] Switching losses occur during both the turn-on and turn-off transitions of a switch.
- [ ] Turn-on loss involves current rising while voltage is high, and then voltage falling while current is high.
- [ ] Turn-off loss involves voltage rising while current is high, and then current falling while voltage is high.
- [ ] Total switching power loss is proportional to switching frequency, blocking voltage, switching current, and the total transition times.
- [ ] Reducing the turn-on and turn-off times of switches is critical for minimizing switching losses.
- [ ] For accurate analysis, the ripple in the load current should be considered, as it affects the current at switching instants.

## Conclusion

The detailed analysis of switching losses underscores a fundamental trade-off in power converter design: the pursuit of miniaturization through high switching frequencies directly conflicts with the goal of high efficiency due to increased

switching losses. This understanding is not merely academic; it forms the bedrock for designing practical, high-performance power converters for applications like charging infrastructure. By quantifying how switching losses depend on factors such as switching frequency, blocking voltage, switching current, and transition times, engineers can make informed decisions about component selection, gate drive design, and overall converter topology. The insights gained from this lecture, particularly the recognition that overlapping voltage and current during transitions are the root cause of these losses, naturally lead to the exploration of advanced modulation strategies and soft-switching techniques. These future discussions will focus on how to manipulate the switching environment to minimize or even eliminate these losses, thereby enabling the development of truly compact, lightweight, and highly efficient power conversion systems.